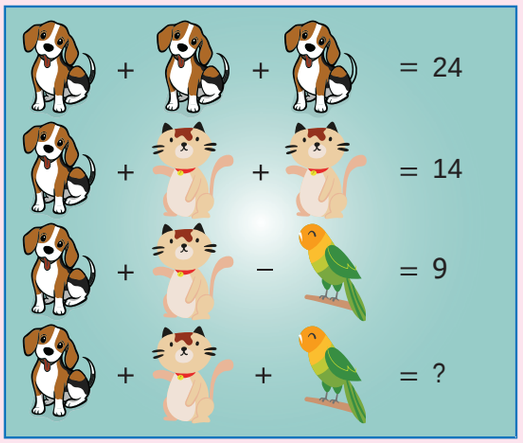
3. An equation remains the same, when the expressions on the left and on the right are interchanged. The equation $7x + 3 = 17$ is the same as $17 = 7x + 3$. Similarly, the equation $7x + 3 = 3x - 1$ is the same as $3x - 1 = 7x + 3$.

Try this

If the dogs, cats and parrots represent unknowns, find them. Substitute each of the values so obtained in the equations and verify the answers.



Example 3.11 Find two consecutive natural numbers whose sum is 75.

Solution The numbers are natural and consecutive. Let the numbers be x and $x + 1$.

Given that,

$$x + (x + 1) = 75$$
$$2x + 1 = 75$$
$$2x + 1 - 1 = 75 - 1 \quad \text{[Subtract 1 on both sides]}$$
$$2x + 0 = 74$$
$$\frac{2x}{2} = \frac{74}{2} \quad \text{[Divide by 2 on both sides]}$$
$$x = 37 \text{ and } x + 1 = 38$$

Therefore, the required numbers are 37 and 38.

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Example 3.12

A person has ₹ 960 in denominations of ₹ 1, ₹ 5 and ₹ 10 notes. The number of notes in each denomination is equal. What is the total number of notes?

Solution Let the number of notes in each denomination be x.

$$\text{Then } x + 5x + 10x = 960$$
$$(1 + 5 + 10)x = 960$$
$$16x = 960$$

Divide by 16 on both the sides,

$$\frac{16x}{16} = \frac{960}{16}$$
$$\text{Therefore, } x = 60$$

Thus, the number of notes in each denomination is 60.

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